Medicine Price Prediction & Recommendation System

1. Project Planning & Management

1.1 Project Proposal

1.1.1 Project Overview

The **Medicine Price Prediction & Recommendation System** is a data-driven healthcare support platform designed to analyze historical medicine pricing data, predict future price trends, and recommend cost-effective alternatives when price increases are expected.

The system aims to reduce the financial burden on patients, particularly those who rely on long-term medication by providing predictive insights that help users make informed purchasing decisions in advance. In addition to price prediction and recommendation capabilities, the system incorporates an intelligent assistant module that allows users to ask medicine-related questions through a conversational interface, improving accessibility and overall user experience.

The project integrates multiple components, including data preprocessing, machine learning prediction models, backend API services, interactive frontend dashboards, visualization analytics, and testing/documentation processes into one unified platform.

1.1.2 Project Objectives

The main objectives of the project are:

- Develop a machine learning model capable of predicting future medicine prices using historical pricing patterns and medicine characteristics.
- Identify potential price spikes and notify users in advance.
- Recommend cheaper alternative medicines based on category similarity and active ingredient matching.
- Provide clear visualizations for medicine price trends and market insights.
- Build a user-friendly interface for easy interaction with the system.
- Implement an intelligent chatbot assistant to answer user questions regarding medicines, pricing trends, and recommendations.
- Ensure system reliability through testing, validation, and performance monitoring.

1.1.3 Project Scope

In Scope

The system will include:

- Medicine price prediction using machine learning models.
- Historical price trend analysis.
- Cheaper alternative medicine recommendation engine.
- Interactive dashboard for visual analytics.
- Backend API for data access and model serving.
- AI conversational assistant for user inquiries.
- Data cleaning and preprocessing pipeline.
- Testing, documentation, and performance evaluation.

Out of Scope

The system will **not** initially include:

- Direct online medicine purchasing.
- Integration with hospital management systems.
- Real-time nationwide pharmacy inventory tracking.
- Prescription generation or medical diagnosis.
- Large-scale commercial deployment infrastructure.

1.2 Project Plan

1.2.1 Project Timeline Overview

The project follows a **hybrid project management methodology**, combining a sequential (Waterfall) structure for academic milestones and an iterative (Agile-style) approach during development phases.

The overall timeline spans from **February 2026 to July 2026**, aligned with university deadlines for deliverables, implementation, and final submission.

1.2.2 Project Phases & Milestones

Phase 1: Project Initiation & Planning (February 2026)

- Define project scope and objectives
- Assign team roles and responsibilities
- Identify tools, technologies, and dataset sources
- Initial feasibility analysis

Milestone: Approved project proposal and planning document

Phase 2: Data Collection & Preparation (February – March 2026)

- Collect medicine dataset (~14,679 records)
- Handle missing values and inconsistencies
- Normalize and clean data
- Feature identification (price, ingredients, manufacturer, etc.)

Milestone: Clean and structured dataset ready for analysis

Phase 3: System Analysis & Design (March – April 2026)

- Define system architecture
- Design database schema (SQLite)
- Create UML diagrams (use case, class, sequence)
- Design API structure and data flow
- Define chatbot retrieval logic design

Milestone: Completed system design documentation

Phase 4: Machine Learning Development (April – May 2026)

- Feature engineering
- Model selection and training (Regression-based models)
- Evaluation using MAE, RMSE, R^2
- Price trend analysis logic
- Alternative recommendation logic using similarity (active ingredients, dosage, category)

Milestone: Trained and validated ML model

Phase 5: Backend Development (May 2026)

- Develop REST API using Flask/FastAPI
- Connect ML model to backend
- Implement database queries (SQLite integration)
- Build endpoints for:
  - price prediction
  - medicine search
  - alternative recommendation
  - chatbot retrieval system

Milestone: Functional backend system

Phase 6: Frontend Development (May – June 2026)

- Build dashboard UI (Streamlit / web interface)
- Display:
  - price trends
  - predictions
  - recommended alternatives
- Integrate chatbot interface
- Improve UX and responsiveness

Milestone: Fully functional user interface

Phase 7: Chatbot Integration (June 2026)

- Implement retrieval-based chatbot system
- Connect chatbot to dataset and backend APIs
- Enable query handling:
  - medicine information
  - price insights
  - alternative suggestions

Milestone: Working intelligent assistant module

Phase 8: Testing & Optimization (June – July 2026)

- Unit testing (backend + ML pipeline)
- Integration testing (full system flow)
- Model performance evaluation
- Bug fixing and optimization

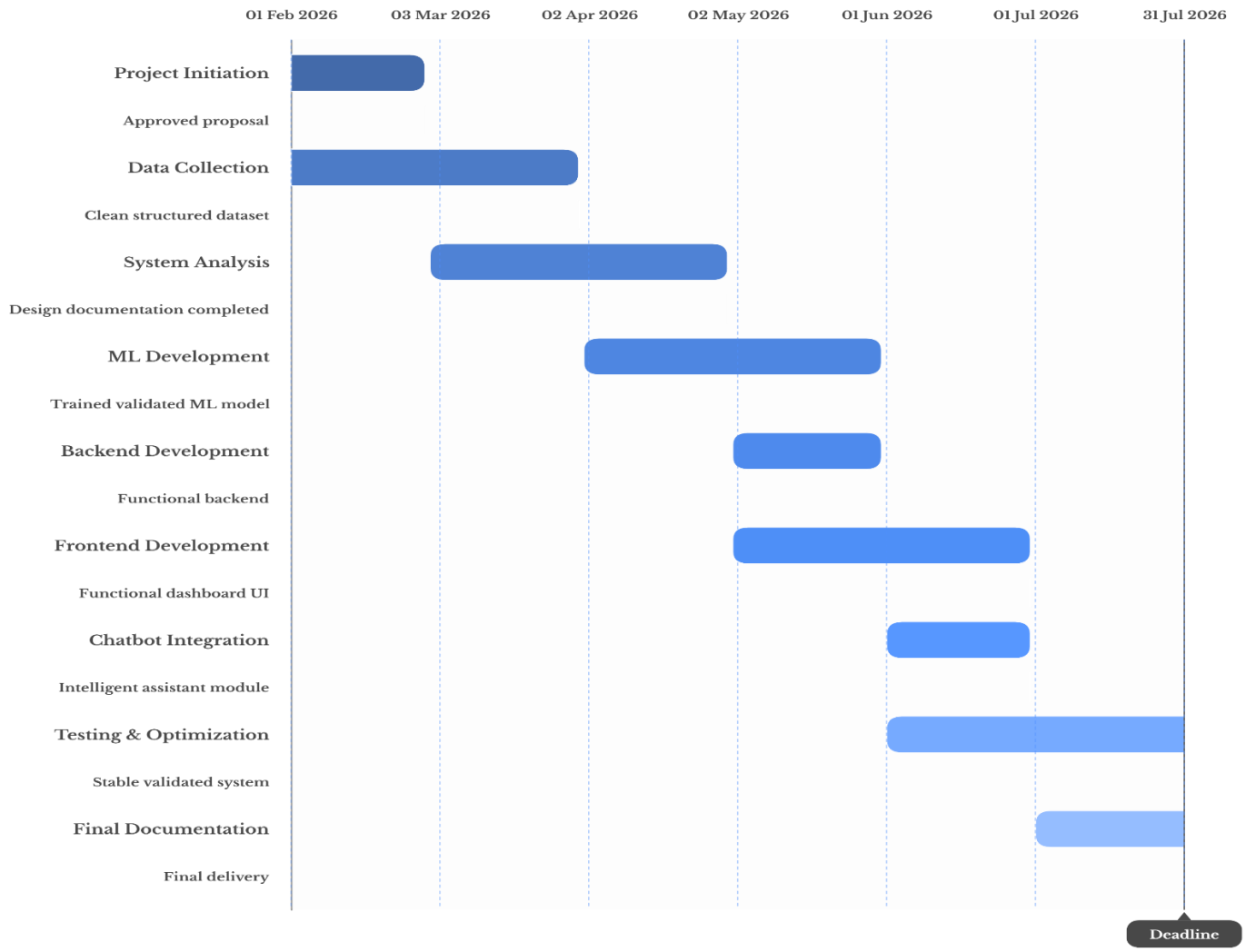
Milestone: Stable and validated system

Phase 9: Final Documentation & Presentation (July 2026)

- Complete project documentation
- Prepare final presentation slides
- System demonstration and report submission

Milestone: Final project delivery

1.2.3 Gantt Chart



1.2.4 Project Management Methodology

The project follows a **Hybrid (Waterfall + Agile) methodology**:

Waterfall Component (Structured Phases)

The project adheres to sequential academic milestones:

- Planning → Data → Design → Implementation → Testing → Delivery
- Each phase must be completed before moving to the next academic milestone

Agile Component (Development Phase)

During implementation phases:

- Backend, frontend, ML, and chatbot modules are developed iteratively
- Continuous testing and refinement are applied
- Features are delivered in incremental cycles rather than all at once
- Collaboration between modules is parallel and flexible

1.2.5 Resource Allocation

Project resources were allocated to ensure efficient execution across all development phases while maintaining balanced workload distribution among team members. The allocation covers human resources, technical resources, and time resources required for project completion.

Human Resources

The project team consists of **six members**, each assigned to a specialized area:

- Data Cleaning & Preprocessing
- Data Visualization & Analytics
- Machine Learning Development
- Frontend Development
- Backend Development
- Documentation, Testing & Quality Assurance

This division enables parallel development of system components and improves overall productivity.

Technical Resources

The project utilizes lightweight and open-source technologies, including:

- Python for development
- Pandas and NumPy for data processing
- scikit-learn for machine learning
- SQLite for database storage
- FastAPI / Flask for backend services
- Streamlit / web technologies for frontend visualization
- GitHub for collaboration and version control

Time Allocation

Project effort is distributed approximately as follows:

Component	Allocation
Data Preparation	20%
Machine Learning	25%
Backend Development	15%
Frontend Development	15%
Visualization	10%
Testing & Documentation	15%

This allocation prioritizes the predictive and recommendation engine while ensuring adequate effort for system development, integration, and quality assurance.

1.2.6 Deliverables

- Clean and structured dataset
- Trained machine learning model
- Backend API system
- Frontend dashboard interface
- Chatbot retrieval system
- System design documentation (UML, ERD, DFD)
- Final project report
- Presentation slides

1.2.7 Key Project Constraints

- Limited dataset completeness (missing values in some fields)
- No real-time pharmacy integration
- No production-level cloud deployment
- Dependency on historical price patterns for prediction

1.3 Task Assignment & Roles

The project team has been organized according to each member's area of responsibility, with defined roles that collectively contribute to the successful development of the **Medicine Price Prediction & Recommendation System**.

The assignment of responsibilities follows a modular development strategy, allowing parallel progress across data engineering, machine learning, backend services, frontend development, visualization, and project quality assurance.

1.3.1 Team Structure Overview

The project team consists of **six members**, each assigned a specialized functional role based on the core components of the system.

Team Member	Assigned Role	Primary Responsibilities
Mohamed Ashraf	Data Collection & Preprocessing Engineer	Data cleaning, missing value handling, normalization, data validation, feature preparation
Anas Tarek	Machine Learning Engineer	Feature engineering, model training, model evaluation, recommendation engine, chatbot retrieval logic
Alaa Osman	Backend Developer	API development, database integration, model serving, endpoint implementation, backend logic
Asem Wael	Frontend Developer	User interface design, dashboard implementation, chatbot interface integration, usability optimization
Haitham Mohamed	Data Visualization & Analytics Engineer	Dashboard analytics, trend visualization, KPI reporting, chart development, exploratory data analysis
Ahmed Tamer	Documentation, Testing & Quality Assurance	Technical documentation, report writing, testing strategy, bug tracking, quality assurance, presentation preparation

1.3.2 Detailed Responsibilities

1) Mohamed Ashraf (Data Collection & Preprocessing Engineer)

Responsible for transforming raw medicine data into structured, high-quality input suitable for machine learning and system use.

Responsibilities include:

- Data collection and consolidation
- Cleaning inconsistent records
- Handling null or incomplete fields
- Data normalization and standardization
- Duplicate record detection and removal
- Feature extraction and transformation
- Data integrity verification
- Preparing processed datasets for modeling and analytics

Key Deliverables:

- Cleaned dataset
- Preprocessing pipeline
- Feature-ready data tables

2) Anas Tarek (Machine Learning Engineer)

Responsible for designing the predictive intelligence layer of the system.

Responsibilities include:

- Feature engineering for predictive modeling
- Model selection and experimentation
- Training predictive models
- Model tuning and validation
- Evaluation using MAE, RMSE, and R^2 metrics
- Developing medicine recommendation logic
- Building similarity matching for cheaper alternatives
- Designing retrieval-based chatbot logic for natural language medicine queries

Key Deliverables:

- Trained predictive model
- Recommendation engine
- Conversational retrieval assistant logic

3)Alaa Osman (Backend Developer)

Responsible for building the core application services and system connectivity.

Responsibilities include:

- Designing backend architecture
- Building RESTful APIs using FastAPI or Flask
- Managing database interactions with SQLite
- Integrating machine learning models into APIs
- Building medicine search endpoints
- Implementing recommendation endpoints
- Connecting chatbot retrieval pipeline
- Ensuring backend security and performance

Key Deliverables:

- Functional backend API
- Database integration layer
- ML service endpoints

4) Asem Wael (Frontend Developer)

Responsible for building the visual and interactive layer of the system.

Responsibilities include:

- Designing system UI/UX
- Implementing dashboard pages
- Building medicine search interface
- Integrating prediction outputs visually
- Creating chatbot interaction interface
- Ensuring responsiveness and accessibility
- Improving usability and navigation flow

Key Deliverables:

- Frontend dashboard
- Chatbot UI
- Interactive search and recommendation pages

5) Haitham Mohamed (Data Visualization & Analytics Engineer)

Responsible for converting raw and processed data into meaningful visual insights that support decision-making and improve usability.

Responsibilities include:

- Building analytical dashboards
- Designing charts for medicine price trends
- Creating KPI visualization modules
- Supporting exploratory data analysis
- Producing comparative medicine pricing visuals
- Monitoring system analytical outputs
- Improving data interpretability for end users

Key Deliverables:

- Interactive dashboard charts
- KPI visualization reports
- Trend analysis visuals

6) Ahmed Tamer (Documentation, Testing & Quality Assurance)

Responsible for ensuring that project outputs are properly documented, validated, and prepared for academic submission.

Responsibilities include:

- Preparing project proposal documentation
- Writing technical reports and system documentation
- Developing testing plans and validation scenarios
- Performing unit, integration, and functional testing
- Tracking defects and coordinating fixes
- Verifying alignment between requirements and implementation
- Preparing presentation materials and final project reporting
- Managing project repository organization on GitHub

Key Deliverables:

- Documentation package
- Testing report
- QA checklist
- Final presentation package

1.3.3 Collaboration Model

Although responsibilities are specialized, project execution follows a collaborative model where components are interconnected:

- **Mohamed → Anas**
  Cleaned data feeds the ML pipeline.
- **Anas → Alaa**
  Trained models are integrated into backend APIs.
- **Alaa → Asem**
  APIs provide services consumed by the frontend dashboard.
- **Haitham → Asem**
  Visualization outputs are integrated into UI dashboards.
- **Ahmed → Entire Team**
  Documentation, validation, testing, and project coordination support all modules.

This collaborative workflow ensures modular development while maintaining system cohesion.

1.4 Risk Assessment & Mitigation Plan

For the **Medicine Price Prediction & Recommendation System**, several technical, operational, and management-related risks were identified during the planning phase. Appropriate mitigation strategies have been proposed to reduce their likelihood and impact.

1.4.1 Risk Identification and Analysis

The following table summarizes the major anticipated project risks, their probability of occurrence, potential impact, and the planned mitigation approach.

Risk ID	Identified Risk	Probability	Impact	Mitigation Strategy
R1	Data quality issues such as missing values, incomplete records, or inconsistent formatting	Medium	High	Apply data cleaning, validation rules, normalization, and preprocessing pipelines before model training
R2	Machine learning model produces weak prediction accuracy	Medium	High	Experiment with multiple algorithms, perform feature engineering, tune hyperparameters, and evaluate models using MAE, RMSE, and R^2

Risk ID	Identified Risk	Probability	Impact	Mitigation Strategy
R3	Difficulty integrating frontend, backend, machine learning model, and chatbot components	Medium	High	Adopt modular development, define clear API contracts early, and perform incremental integration testing
R4	Project schedule delays due to workload imbalance or technical challenges	Medium	High	Follow milestone-based planning, track weekly progress, and redistribute tasks when necessary
R5	Dependency bottlenecks where one module delays another module's progress	Medium	Medium	Develop modules in parallel where possible and define fallback mock data/services during integration
R6	Scope expansion caused by adding unnecessary features late in development	Low	Medium	Maintain focus on core deliverables and evaluate any additional feature based on feasibility and time constraints
R7	Retrieval-based chatbot provides irrelevant or low-quality responses	Medium	Medium	Improve indexing and filtering logic, test common user queries, and refine retrieval matching criteria
R8	Insufficient testing may lead to hidden bugs or unstable system behavior	Medium	High	Implement structured testing plans including unit testing, integration testing, and user acceptance testing

1.4.2 Risk Mitigation Strategy

To minimize project uncertainty, the team adopts a **proactive risk management approach** rather than responding only after issues occur. This includes:

- continuous monitoring of project milestones,
- early technical validation of major modules,
- modular development to reduce integration complexity,
- frequent internal testing during implementation,
- maintaining clear communication among team members,
- and prioritizing core project functionality over optional feature expansion.

Special attention is given to **data quality**, **model performance**, and **system integration**, as these represent the most critical technical risks affecting project success.

1.4.3 Contingency Planning

If major risks materialize, the following contingency actions will be taken:

- If model performance is below acceptable levels, fallback models with simpler but stable performance will be used.
- If chatbot retrieval quality is weak, chatbot functionality may be limited to curated medicine information and recommendation support only.
- If integration delays occur, individual modules may be demonstrated independently while final integration is completed incrementally.
- If schedule pressure increases, optional features will be postponed in favor of completing core system functionality.

This contingency planning ensures that the project remains achievable even under technical or scheduling constraints.

1.5 Key Performance Indicators (KPIs)

The selected KPIs for this project are aligned with the system's core components, including machine learning prediction, recommendation quality, backend performance, user experience, and overall system reliability.

1.5.1 Performance Metrics

KPI	Description	Target
Prediction Accuracy	Measures how closely predicted medicine prices align with actual observed prices	≥ 80% acceptable model performance
Mean Absolute Error (MAE)	Average absolute difference between predicted and actual prices	As low as reasonably achievable
Root Mean Squared Error (RMSE)	Measures magnitude of prediction error while penalizing larger deviations	Lower than baseline model
Model Stability	Consistency of model performance across validation datasets	Minimal variation between runs

These indicators evaluate the effectiveness of the machine learning component, which is the core intelligence layer of the system.

1.5.2 System Performance Metrics

KPI	Description	Target
API Response Time	Average time required for backend API requests	< 2 seconds
Dashboard Load Time	Time required for frontend dashboard components to render	< 3 seconds
Chatbot Response Time	Average response generation time for user queries	< 3 seconds
System Availability	Percentage of time the system remains operational during testing	≥ 95%

These metrics ensure that the system remains responsive, accessible, and practical for real-world usage.

1.5.3 Recommendation & Chatbot Quality Metrics

KPI	Description	Target
Recommendation Relevance	Degree to which recommended alternatives match medicine category and active ingredients	High similarity relevance
Cost Reduction Opportunity	Percentage of recommendations that offer a lower-cost alternative	Positive measurable reduction
Chatbot Retrieval Accuracy	Ability of chatbot to return relevant medicine information	≥ 80% relevant response rate
User Query Coverage	Percentage of common medicine-related questions successfully handled	Broad coverage of expected queries

These indicators measure how useful the recommendation and conversational assistant modules are to end users.

1.5.4 Quality Assurance Metrics

KPI	Description	Target
Bug Count	Number of unresolved major defects before final submission	Minimal critical bugs
Integration Success Rate	Successful communication between frontend, backend, database, and ML components	≥ 95% successful transactions
Testing Coverage	Percentage of major modules tested	Full core module coverage
Documentation Completeness	Degree of completion of technical and user documentation	100% submission-ready

These metrics ensure project stability, maintainability, and readiness for deployment or demonstration.

1.5.5 Success Criteria

The project will be considered successful if it achieves the following:

- Produces reliable medicine price predictions with acceptable error rates.
- Recommends relevant lower-cost medicine alternatives.
- Provides a responsive dashboard and conversational assistant experience.
- Maintains stable integration across all system components.
- Completes testing, documentation, and final delivery according to project deadlines.

By monitoring these KPIs throughout development and testing, the team can objectively evaluate project progress and final system effectiveness.